Task 1- AI Problem Design AI Model Selector System

1. User

The main users of the AI Model Selector System are developers, students, researchers, and teams who need to choose an AI model for a particular task. The system is useful when a user is not sure which model will be suitable for the required task, budget, speed, accuracy, and deployment requirements.

2. AI Problem

There are many AI models available for different tasks. Each model can be different in terms of accuracy, speed, cost, cloud provider, region, and availability. Because of this, selecting the right model can take time and may lead to choosing a model that does not match the actual requirement.

The problem addressed in this project is to design an AI-based system that takes the user's requirements and recommends a suitable AI model.

The user can provide requirements such as:

- Type of task
- Required accuracy
- Required speed
- Budget or cost preference
- Cloud provider
- Required region
- Model availability

The system then gives a suitable model recommendation along with the reason for the recommendation.

3. Data Source

A small structured dataset is used for the model selection problem. It contains information about different AI models and their important characteristics. The information is organized so that the system can compare the user's requirements with the available models.

Data	**Description**
Model name	Name of the AI model
Task	Type of task supported by the model
Accuracy	Relative accuracy or performance information
Speed	Relative response speed
Cost	Cost or cost level
Cloud	Cloud/provider information
Region	Supported region information
Availability	Whether the model is available for use

For evaluation, different user requirement cases can be created from these factors. Each case contains an expected suitable model so that the system's recommendation can be compared with the expected result.

4. Constraints

1. The dataset is small and contains a limited number of AI models.
2. The recommendation depends on the quality and correctness of the model information.
3. Model price and availability can change with time and may depend on the region.
4. The system considers only the factors included in the dataset and user input.
5. The system is designed for the selected AI tasks and may not cover every possible AI use case.
6. The recommendation is a decision-support result and is not a guarantee that the selected model will always be the best choice.

5. Evaluation Approach

The system will be evaluated by checking how well it ranks the available AI models according to the user's requirements. For each user requirement, the system compares the available models based on factors such as task, accuracy, speed, cost, cloud, region, and availability. The system then provides the models in order of suitability.

For example:

Rank	Model	Suitability
1	Model A	Best suitable
2	Model B	Second best
3	Model C	Third best

The first model should be the one that matches the user's requirements the best. The next models should be ranked according to how closely they match the requirements.

The evaluation will check:

- Best model selection: Whether the most suitable model is placed at Rank 1.
- Ranking order: Whether the other suitable models are placed in the correct order based on their suitability.
- Requirement matching: Whether the ranking considers the user's important requirements such as cost, speed, accuracy, cloud, and availability.
- Response time: How much time the system takes to generate the ranked list.

Success Criteria

Measure	Target
Best model at Rank 1	At least 80% of test cases
Correct model within Top 3	At least 90% of test cases
Important requirement matching	At least 85%
Average response time	Within 3 seconds

The system will be considered successful if it consistently places the most suitable model first and provides other suitable models in a meaningful order.x